Certified Rewrite Discovery for Tensor Programs: A Machine-Minted Conv+BN Fold from Minting to Runtime

Draft — proposed, not accepted

mlopt project

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Abstract

Compilers choose better implementations only from transformations a human already wrote. We describe a pipeline in which the rewrite rules themselves are a product of measurement: laws are data, composites are minted by a fixpoint over the shipped library, guards are transported through an evidence store, derivability is credited against an object's own premise closure, and admission is decided by a staged gauntlet that has repeatedly falsified the engine's own candidates. The flagship result is `conv_bn_fold` — the eval-mode batch-norm weight fold a hand optimizer writes — minted, admitted, and shipped verbatim as a machine-authored law in the default rule set. The deployed program evaluates bitwise-equal to the spelled one through an independent runtime and is ~5% cheaper per call in the honest deployed regime — a −16% subprocess arm conflates unused marshal work into the spelled baseline and is quoted only in the regime split — while the term-level cost model prices the fold at −33% on the shipped term (−50% on the probe site). All reported numbers are reproducible across the `handl-lang` trees at the locked revisions (§Section 8). The scope is deliberately honest: a kernel-scale, certified-rewrite result, not a model-scale speedup claim.

> **!** Important
>
> **Status.** This is a working draft — *proposed, not accepted*. It has not had human revision or external review. Every quantitative claim names the probe, test, or retro that produced it; the rest of the paper should be read as a candidate for rewriting, not as a settled result.

1 Introduction

Every optimization a production compiler knows was written by a person — a hand-derived rule, encoded per framework. The rule library is the bottleneck: human authorship is slow, and the pairings nobody names never get found. For a machine to *author* optimizations, four things must exist: rules it can write (data, not code), a referee to check them, a score to rank them, and a world — a runtime — that programs, rules, and derivations all live inside.

The pieces have arrived separately. Massalin's superoptimizer (Massalin 1987) searched programs exhaustively; equality saturation (Tate et al. 2009; Willsey et al. 2021) closes the program space under a fixed law bank, and TENSAT applies it to tensor graphs. Rule-inference work (Ruler (Nandi et al. 2021), the Babble line (Cao et al. 2023)) grows the library itself. And learned play on well-defined games — AlphaGo Zero (Silver et al. 2017), AlphaTensor (Fawzi et al. 2022) — reaches

structure beyond human-written heuristics. The synthesis story for *tensor programs with runtime-priced deployment*, though, is thinner: no system connects discovery, certification, pricing, shipping, and learned guidance inside one substrate.

This paper reports the connection — and one complete measured traversal of it — in the mlopt engine. The contribution is not any single mechanism but their composition: a categorical substrate makes rules data, derivations checkable, cost measurable, and provides the runtime world — which closes discovery into a well-defined game played by learned policies. The enabling property is *self-containedness*: programs, rewrite laws, and derivations are n-morphisms in one calculus — executable values, not external metadata — so the system produces the values it consumes. To our knowledge this is the first *operational* substrate of this kind; categorical languages prior to it are denotational or design-level, not working machines whose runtime executes the morphism hierarchy. The claim is narrow and checkable:

> A rewrite a hand optimizer would write — the eval-mode conv+BN weight fold — was **discovered, certified, priced, and shipped by the machine**, and its deployed form runs bitwise-equal and cheaper through an independent runtime.

No model-scale speedup is claimed. The result is kernel-scale (one op pair, fp64 fixtures), the runtime is a tree-walk interpreter (debug build), and the move alphabet is bounded by the 28-op host-tensor surface — all named in Section 7. What the case study establishes is the *pipeline*: that a law can travel from measured need to a shipped, certified, runtime-priced rule without a human writing it.

The repository is `handl-lang/graphopt`; every number below cites the artifact that produced it (§Section 8).

2 Background

2.1 Equality saturation

An e-graph compactly represents the closure of a term under a law bank — all reachable spellings, sharing subterms, to a fixed point (Tate et al. 2009) — and extraction picks the cheapest member under a cost model. The machinery lives in `handl-lang/eqsat` and the platform paper (mlopt project 2026) describes it once: the term IR, the bounded saturation, the certificate and cost algebra. This paper recounts only what the discovery story uses. One piece does carry over verbatim: the soundness boundary is the law bank — a rule's side conditions are the *only* layer that keeps a false member out, and the certificate replay re-proves that a step is a valid application, not that the rule is true.

2.2 Laws as data on a categorical substrate

Catopt reads a program as a 1-morphism between tensor types, a rewrite law as a 2-cell between programs, and a derivation — one law proven from premises — as a 3-cell: the cells-as-data pattern the kernel calculus defines (handl project 2026) and the eqsat paper operationalises (mlopt project 2026). The consequence is that laws are **data rows** (a pattern pair plus a `cond`/`check`/`derive` guard spec), serializable, storable, composable — and what this paper mutates is the bank: `mlopt_core.laws.ALL_RULES` mixes human-authored axioms with 59 machine-admitted objects (`mlopt_core.laws.provenance.MACHINE_SHIPPED` — 13 promoted laws, the 23 minted `prim:synop_*/synop2_*` folds seated in `DEFAULT`, and their 23 `.expand` definitional premises).

2.3 The handl substrate and mlopt-kit

The same "one language, many interpretations" model is implemented by the handl platform — the closed arrow calculus and `Language` schema are the kernel paper's subject (handl project 2026): eval, typecheck and rewrite are interpretations of one language whose tables are data. `packages/mlopt-kit` — a plain Rust crate — composes handl's canonical `Language`, its lex/parse/tree-walk-exec engines, and `handl-host-tensor`'s 28-op `tensor.*` surface as dependencies (no forks). `hnl-run` is a subprocess door: `.hnl` in, a JSON wire tensor out, `--repeat N` evaluating a source N times in one process — the measurement hook. `bench/drivers/kit/term_hnl.py` renders any mlopt IR term as a `.hnl` program from a data table (per-op arg assembly, attr marshaling, leaf binding), so the *same object the e-graph rewrites* is what the runtime executes.

3 Machinery

3.1 The mint fixpoint

`mlopt_core.meta.synthesize_rules` runs the substrate's bounded mint fixpoint (mlopt project 2026) over the law bank on a seed term — apply the rules, compose the pairs that chain, emit `syn_*` composites carrying premise provenance. On the eval-mode ConvBN seed it emits 974 rules in ≈ 12 s (fuel 30,000, 4 rounds); the fold composite mints deterministically (identical names across runs).

3.2 Guard transport

A minted rule's side conditions are closures over its premise objects, so the evidence store keeps them as *specs*, not code — parent names plus re-expression maps — and `evidence.admit_object` rebuilds `check`/`derive` by resolving parents transitively; the substrate's shipped form (`mlopt_core.laws.minted`) embeds the records and replays the rebuild at import, refusing a degraded parent (mlopt project 2026).

3.3 Premise-anchored derivability

`verify_law` answers "is this instance derivable under a rule bank?" with a replayable witness, and for a stored composite the honest question — derivability under the premises it was minted from — is the substrate's anti-gaming discipline (mlopt project 2026): the certificate must be standalone-replayable and every step naming a non-axiom rule must *site-verify* under the fp64 numeric oracle. A premise false at its applied site revokes the credit and the verdict falls back to the axioms-only answer.

3.4 One cost alphabet

The substrate's cost model credits compile-time work structurally (mlopt project 2026): a subtree closed over `Param`/`Const` leaves whose root op is in `FOLD_OPS` — the 161-op fold-at-lowering alphabet — is priced at ~0 marginal. This is what lets a fold's pay leg measure what it should: the fold's `mul`/`reshape`/`rsqrt` chain is once-per-deployment work, not per-call work, and `dag_cost` prices it that way.

3.5 The admission gauntlet

`evidence.run_gauntlet` is the domain-side staging of the substrate's admission protocol (mlopt project 2026): reconstruct → full-data → measure → truth → novelty → typed-pay → closure →

certificate, with truth the two-legged numeric-oracle-plus-derivability check and novelty refusing respellings of shipped laws (`relation=duplicate`). What this paper adds is the record of it biting: the gauntlet has repeatedly falsified the engine's own candidates — a naive broadcast guard with 212 bad accepts, an apparent paying sibling with semantically unequal sites, four deliberate false twins (`project/retros/mul-unsq-nat.md`, `layer3-loop.md`, `typed-pay-gate.md`).

3.6 Learned guidance as a game

Because laws are data, discovery admits a well-defined game structure — moves are rule applications, the admission gauntlet is the referee, the cost alphabet is the score (mlopt project 2026). This is implemented machinery, not aspiration, and it lives in `handl-lang/gamekit`: `gamekit.engine` defines the `Board` protocol and the episode/train/eval drivers, and `gamekit.players` provides the `LinearPolicy` arm over a domain-specified featurizer — `mlopt_discovery.engine/players` are compatibility re-export shims over those modules. `mlopt_discovery.play` carries the domain registry (`meta`, `joint`, `torch`, `search`, `gen`, `open`, `player`; `torch` and `gen` ship `deliver` hooks that lower the winning extraction to a verified module). A domain is data — a case generator, a board factory, and a feature schema — not a subclass.

Two linear priors ship with pinned held-out wins: the open-game policy ranks law candidates at ~4.7× the cost-delta heuristic's average precision (AP 0.210 vs 0.045) and ~2× useful laws per episode, and the meta-arena prior beats the scripted playbook (mean episode reward −0.06 vs −0.10) — the latter flagged by its own eval as possibly sampling-variance-riding, an honest negative kept in the record. A distilled contraction policy (`PairPolicyNet`) beats `opt_einsum`'s randomized greedy (0.91× cost at n=40) and ships as a standalone artifact. The neural arm's residual gap is ordering credit, not coverage: a pairwise RankNet objective on the reward's own label reaches AP 0.1941 — above the trained-linear anchor (0.128), under the page-aware linear (0.210) — and a ListNet-graded drain earns heuristic-parity episode reward (0.3549 vs 0.3595, inside seed noise) without beating it (`docs/results_neural_rank.md`). Two learned-policy axes were measured and falsified: learned *schedule guidance* over the generator arms (`project/retros/guide-real-run.md`, `guide-learned-corpus.md` — the optimal schedule reduces to a rule), and *direct law invention* by a learned policy (~10× worse than structured construction; ADR 0004). The evidence for learned guidance is real but early (controlled baseline and ablations in `docs/results_learned_baseline.md` / `docs/results_learned_ablations.md`; dedicated RL arms in `docs/results_learned_rl.md`): linear and distilled arms beat baselines; neural methods reach parity, not wins.

4 Case study: the conv+BN fold

The rewrite is the standard eval-mode fold:

$$\mathrm{BN}_{\gamma,\beta,\mu,\sigma^2,\varepsilon}(\mathrm{conv}(x,W,b)) \;=\; \mathrm{conv}(x,\; W \odot \hat{s}',\; b \odot s + (\beta - s \odot \mu)), \qquad s = \gamma \cdot (\sigma^2+\varepsilon)^{-1/2}.$$

The account below is the probe's own output (`bench/drivers/probes/conv_fold_premise_pay_probe.py`, ~13 s).

Anchor. `CONV_FOLD_RULES` provides the channel-scale and channel-shift anchors (`mul(conv2d(x,W,b),ŝ)`-shaped single-step folds); `BATCH_NORM_RULES` expands `batch_norm` into its affine spelling. Individually they are enabled-but-opt-in — the anchors refuse at truth-starvation, the expand at typed-pay — their value is *enablement*: they are the fold's derivation premises.

Mint. `synthesize_rules` over the bank minus the already-shipped composite emits 974 rules; the fold composite — `batch_norm(conv2d(·),…)` → `conv2d(·)` — mints as a depth-4 `syn_*` chain over an 8-rule transitive premise closure (the `batch_norm_expand` expansion + four `conv_channel_*` anchors + three `syn_*` intermediates).

Transport and truth. Stored with its closure, `admit_object` rebuilds both hooks from guard records (`missing=('check','derive')` → transported). The transported guard accepts the real site (1 eq / 0 ne) and 105/105 of the guarded synth region with 255 honest declines; `verify_law` replays the derivation under shipped axioms in one step — `witness=('conv_bn_fold',)`, the library law proving its own minted origin.

Pay. On the 8×8 probe site the folded instance drops `dag_cost` **34,874.3 → 17,438.7** (−50 %), fires once, verifies `pass` on the lowered modules.

Ship. The composite is promoted *verbatim*: its stored records embed in `mlopt_core.laws.minted`, its guard re-materializes at import, `MACHINE_SHIPPED` grows to 12 (59 today — `add_unsq_pad_r`, then the primitive arm's 23 folds and 23 `.expand` premises followed). Re-running the gauntlet post-ship refuses at novelty with `relation=duplicate` — the documented signature of "already admitted."

Runtime-verified. Through the kit bridge (Section 2), the spelled term `batch_norm(conv2d(x,W,b),…)` and the deployed term `conv2d(x,W2,b2)` — with `W2`,`b2` themselves mlopt subterms materialized through the kit — evaluate to the same tensor (**bitwise-equal** on the seeded fixture; the full `torch.export` IR of the module evaluates at 6.7e-16 max abs diff), and per-call wall over `hnl-run --repeat 30` measures three regimes — the honest deployed arm (fold math amortized to load) is ∼−5 %; an inline-fold arm that recomputes the fold per call runs *slower* (+22–33 %); and a subprocess-arm −16 % figure conflates the prelude's unused fold-marshal work into the baseline. `dag_cost` is the stable headline: 55,296 → 36,864 (−33 %). Full reconciliation: `bench/RESULTS.md` + `bench/drivers/probes/conv_fold_wall_regimes_probe.py`.

: The fold decision, priced by the runtime itself — honest regime-split. {#tbl-decision}

One honest subtlety: `dag_cost`'s param-only discount models compile-time folding the tree-walk runtime does not perform, so the measured arm separates fold-once (materialize `W2`,`b2`) from run-every-call (the deployed conv). The cost model and the wall agree in direction; the wall numbers are debug-build timings — directional, not a benchmark rig.

5 Frontier measurements

The discovery loop's own frontier was measured this cycle as a pre-registered question series (`project/TODO.md`). The verdicts sharpen where the open work lives — and it is not where the case study would suggest.

arm	question	measured verdict
Q1	unnamed structure on *real* corpora	**rewrites saturate at known-art; primitives break it** — the rewrite mint over 53 held-out zoo modules minted 102 viable candidates and admitted 8 paying folds, all **known-art** (already catalogued); the *primitive* arm, which mints new `synop_*` ops for recurring manual shapes no bank op denotes, produced on the 68-module corpus **21/24 admitted machine-original folds** at depth 1 — certified, paying (launch 0.36–0.80x), novel by construction, now seated in `DEFAULT` — then **48/48** on the widened `--prim-deep` hunt and **2 depth-2 `synop2_*` folds** under `--prim-bootstrap` (shipped folds seated as vocabulary); **71 distinct machine-originals across runs**. The honest content: it recovered the shared gated-accumulator skeleton of the SSM family (DeltaNet/GLA/Mamba/RG-LRU) — real cross-architecture structure, not unknown-to-humanity transforms (`docs/reports/primitives_finds.md`)
Q2	rediscovery of a Strassen-class algorithm	**NO-GO** — the 2×2×2 bilinear board provably expresses rank 7 (a scripted 29-move line verifies 10/10), yet random, greedy-repair, and cold-linear players stalled at rank 8 over 200+ episodes at budgets 48 and 96 (`docs/results_strassen.md`)

arm	question	measured verdict
Q3	multi-step credit — mint a premise, cash it later	**YES, where chains are dense** — the trained linear arm argmaxes the enabling chain on 100% of held-out boards and the neural arm converges to the scripted ceiling; at sparser chain density (more decoys, fewer episodes) linear REINFORCE degrades to the myopic arm (`docs/results_credit_chain.md`)
Q4	scale	**three distinct walls** — matcher time on unrolled terms (~11k IR nodes exceeds the 480 s rung budget), verify-forward RAM on the *spelled* program (a ~17 GB fp64 attention tensor at T=4096), and spelled-program numerics (fp64 overflow; verify honestly refuses at T 1024) (`docs/reports/results_scale_ladder.md`)
Q5	prior transfer to a second domain	**null** — a private codegen-domain measurement, cited here without numbers

One substrate bug surfaced en route and is fixed: emit-time RHS sanity conflated *cannot-evaluate* with *evaluated-and-false* through a bare exception catch, so non-denoting composites were being minted as candidates (eqsat `2d6e4eb`). The mint now emits only viable objects — the 102-mint funnel above is the post-fix count.

The pair that teaches. Q2 fails where Q3 succeeds, and the difference is not the machinery — the same board protocol, players, and certifying referee appear in both. It is the shape of the productive trajectory: an enabling chain is a short mint → apply pair the episode stream generates often enough to learn from, while the naive → Strassen path is a ~15-move coordinated sequence through invalid-space intermediates with no positive prefix. **Search depth and chain density — not expressibility, not the referee — are the frontier.**

6 Related work

Equality saturation and e-graph superoptimization. Tate et al. (Tate et al. 2009) introduced the technique; egg (Willsey et al. 2021) made it practical; TENSAT (Yang et al. 2021) and follow-ons apply it to tensor graphs with fixed, hand-written rule sets. Catopt uses the same substrate but the rule bank is not fixed — the discovery loop writes into it through the gauntlet. egglog (Zhang et al. 2023) serves as a test-only differential oracle for the hand-rolled saturation.

Rewrite-rule inference and library learning. Ruler (Nandi et al. 2021) infers rewrite rules

for scalar domains by enumeration + equality-saturation filtering — the direct ancestor of the mint-then-verify shape here, differing in that mlopt mints *composites over tensor laws* with transported guards and credits derivability against an object's own premise closure. Babble (Cao et al. 2023) learns library abstractions by anti-unification; mlopt's definitional-atom layer is the same move and measured **null** on reach-expansion — the paying layer is above it.

Generated rule banks. TASO (Jia et al. 2019) auto-generates graph substitutions for DNNs and verifies them formally with an automated theorem prover — the strongest prior on the verify-what-you-mint leg; the generated bank ships once and does not re-enter subsequent discovery. Isaria (Thomas and Bornholt 2024) synthesizes *sound* rewrite-rule systems from ISA specifications to drive e-graph vectorizing compilers, and Enumo (Pal et al. 2023) makes rule inference programmable — both grow the bank offline, and the ruleset a compiler runs is static. Glenside (Smith et al. 2021) is the closest precedent for machine-derived tensor transformations: e-graphs rediscovering access-pattern rewrites (im2col-style) — the shape of the conv+BN fold here.

Learned play over rewriting. CompilerGym (Cummins et al. 2022) casts compiler optimization as an RL environment over a fixed action space; the Cambridge line — Omelette (Singh 2022), MCTS-GEB (He, Singh, and Yoneki 2023), Aurora (Bărbulescu et al. 2024) — trains RL/MCTS policies that steer e-graph construction. Learned play over rewriting exists, but over a static bank. AlphaGeometry (Trinh et al. 2024) couples a learned proposer to a certifying symbolic deducer — learned play inside a verified loop, over a fixed deduction closure (AlphaTensor (Fawzi et al. 2022) similarly searches one fixed structure).

Each neighbor holds pieces of the loop — generation, certification, learned play — but, to our knowledge, none makes the minted bank itself an object of play: in mlopt a law minted during discovery re-enters as a move inside the learned game (Section 3.6). The differentiation is that closure, not any single leg.

Three capabilities fall out of the closure and belong to no single neighbor. *Two-engine certification*: a certified law must execute identically through mlopt's e-graph *and* through handl's independently-implemented rewrite engine (`packages/mlopt-handl-peer`; the audit driver `bench/drivers/audit/handl_differential.py` fires the structural bank on both and reports disagreement) — a wrong law must fool both. *Laws as first-class values*: minting constructs a `Value` in the same calculus the engine executes — a data row, not generated code — so admission is a runtime decision, not a build step. *Language-agnostic exits*: the `Sink` port lowers one extracted graph to torch (`mlopt_torch.TorchSink`) or ONNX (`mlopt_onnx.OnnxSink`), and a separate bench driver (`bench/drivers/kit/term_hnl.py`) renders the same term as handl `.hnl` — so a certified result is not married to the backend that priced it.

Compilers. TVM (Chen et al. 2018) autotunes schedules; TorchInductor (Ansel et al. 2024) fuses pointwise ops and codegen's the graph a human optimizer could reach. Both are incumbents in the repo's benchmark tables (wins and losses alike); the contribution here is not a faster kernel but a certified, machine-discovered *rewrite*.

Certified/verified compilation. The per-artifact boundary design — law, search, delivery, bound each with its own verifier — is in the spirit of translation validation; what is unusual is that the verifier also audits the *discovery* machinery (adversarial sweeps falsifying minted guards).

7 Limitations

- **Kernel scale.** One op pair, fp64 fixtures, a 1×4×24×24 activation. Nothing here measures model-scale compilation; wider-corpus evidence in the repo (bimodal exposure, ~1.5% yield of fresh paying structure) is honest that this is not yet a general speedup.
- **Move alphabet bound.** The kit surface is 28 `tensor.*` ops (16 forwards + 11 `*_backward` adjoints + `derive`); the fold's own $s = \gamma \cdot \mathrm{rsqrt}(\sigma^2 + \varepsilon)$ is outside it (no elementwise `rsqrt`) and rides a marshaling channel. Emitted terms are confined to what the surface spells; outside-surface ops refuse by name.
- **Tree-walk runtime.** `hnl-run --repeat` timings are wall-clock on a debug-build tree-walk interpreter — they establish direction (the fold's value lands in the deployed program), not competitive throughput. No codegen claim is made.
- **Single-case gauntlet.** The admission measured here is a one-term corpus; the machinery's broader soundness evidence lives in the library-level sweeps, which caught real holes — but on different objects than this one.
- **Attr-narrow LHS.** The shipped composite's LHS carries the seed's literal attrs (`stride=(1,1)`, `padding=(1,1)`, `eps=1e-05`, `training=False`): its region is the default-attr eval-mode fold — honest, narrow, the common spelling.
- **Novelty is closed-world.** "Novel" in this draft means *machine-original against catalogued knowledge* — the decidable cert ∧ beat ∧ ¬derivable predicate the platform paper defines (mlopt project 2026), evaluated against `known_bank()` — the shipped rules plus the `prior_art` preset's `FUSION/CATEGORICAL` rows (the slice the bounded prover can saturate against) — minus alpha-siblings. The sieve (`bench/drivers/screen/novelty_sieve.py`, results in `docs/results_novelty_sieve.md`) ran 24 candidates over the then-242-rule bank (`known_bank()` now counts 265 — the shipped `prim:synop*` folds seated since): the review queue is **empty** — the two earlier review items (`mul_unsq_nat`, `softsign_fold`) demoted to derivable once their catalogued duals (`mul_unsq_rev`, `softsign_expand`) were registered as prior art — six derivable, six unpaid, twelve unmeasured. "Novel to humanity" remains an open-world literature claim this draft does not make.
- **Scale walls are measured, not hypothetical.** Saturation is T-invariant on symbolic attention terms ($\lesssim$ 0.3 s through T=4096), but unrolled spellings hit a matcher wall around ~11k IR nodes (T=1024 exceeds a 480 s budget), the memory wall is the referee's verify-forward of the spelled program (~17 GB at T=4096 — the e-graph had saturated that term in 0.2 s), and an unscaled `exp` attention overflows fp64 at T $\geq$ 1024 — verify correctly refuses rather than certify a NaN (`docs/reports/results_scale_ladder.md`).
- **Rewrite-space originality on real corpora is closed; primitive space is open.** The rewrite mint admitted 8 paying folds — every one already a theorem of the registered prior-art bank (known-art). The primitive arm's machine-original folds (21 at depth 1, 48 in the widened `--prim-deep` run, 2 at depth 2 under `--prim-bootstrap` — 71 distinct across runs) are novel *by construction* (the `synop_*` op did not exist), which is exactly the falsifiable claim — but the minted shapes are recognizable structure (mul-add trees, broadcast select-gates), not transforms a compiler engineer would call new. Whether a minted primitive could acquire a fused *implementation* the machine also found is answered, twice over: the v4 `gather` handler family mints verified bodies for **24/24** (`docs/reports/prim_handlers.md`), and `family_synth.mint_families` recovered `gather` itself as a minted descriptor (`fam_8c9a7ed8`, parity-verified — the family vocabulary is now an object of the mint, `docs/reports/family_synth.md`); the residual open question is whether a minted body is a kernel a fused-kernel engineer would call *new*. Strassen-class rediscovery

remains out of reach of per-step play (Section 5).

8 Reproduction

Reproducibility is org-scoped, stated honestly. The flagship artifacts — the `conv_bn_fold` shipped law, `MACHINE_SHIPPED` provenance, the `dag_cost` alphabet — live in `mlopt_core` in `handl-lang/eqsat`, and the game layer in `handl-lang/gamekit`; both resolve as private `git+ssh` dependencies pinned in `uv.lock` (eqsat `194b305e…`, gamekit `450731f…`). The measured retros each claim cites live under `project/retros/`, which is gitignored on `main` — they are committed on the repo's `project` branch, not in a `main` clone. So: reproducible across the `handl-lang/{graphopt,eqsat,gamekit}` trees at the locked revisions, requiring org access — not from this repo alone.

Within that scope:

```
uv sync
.venv/bin/python bench/drivers/probes/conv_fold_premise_pay_probe.py   # the admission account
.venv/bin/python -m pytest tests/test_kit_bridge.py -q       # the runtime-priced decision (8 t
cargo test -p mlopt-kit                                     # the Rust consumer (31 tests)
.venv/bin/python -m pytest -m "not slow"                     # the dev lane
```

The evidence ledger is `bench/RESULTS.md`; the measured retros each claim cites are under `project/retros/` on the `project` branch (notably `minted-composite-default.md`, `kit-term-bridge.md`, `premise-anchored-derivability.md`, `fold-cost-single-source.md`); the extended working draft with the full win/loss tables is `docs/paper_draft.md`.

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